Randomized and low-rank approximation

Constant-factor HSS approximation in polynomial time

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because shared HSS bases couple approximation choices across levels; community impact is reliable compression for hierarchical numerical solvers.

Area: structured low-rank matrix approximation

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^{\mathsf{T}}v$ for one chosen real vector v; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm's randomness.

For $N = 2^{L+1}k$, $k, L \geq 1$, form the perfect binary tree that successively halves the ordered index set $[N] = \{1, \dots, N\}$ until every leaf contains $2k$ indices. Define $\mathcal{S}_{L,k}$ to contain precisely the real $N \times N$ matrices B such that, for every nonroot node I of this tree,

$$\operatorname{rank} B[I, [N] \setminus I] \leq k, \qquad \operatorname{rank} B[[N] \setminus I, I] \leq k.$$

This is the HSS class with rank at most k, including its constraints across levels. There is no restriction inside a leaf diagonal block. Set

$$E_{L,k}(A) = \min_{B \in \mathcal{S}_{L,k}} \|A - B\|_F^2.$$

The minimum exists. The rank definition follows Stefano Massei, Leonardo Robol, and Daniel Kressner, *hm-toolbox: MATLAB Software for HODLR and HSS Matrices*, SIAM Journal on Scientific Computing **42** (2020), C43–C68, §2.2, Definition 3. Equivalence with the telescoping representation and existence of a minimizer are recorded in Amsel et al., cited below, Definition 3 and Appendix D.

Problem statement

Do absolute constants $C, c > 0$ and a uniform randomized algorithm exist that, given the entries of any $A \in \mathbb{R}^{N \times N}$ and integers $k, L \geq 1$ with $N = 2^{L+1}k$, returns $B \in \mathcal{S}_{L,k}$ using $O(N^c)$ arithmetic operations and satisfies

$$\mathbb{E}\|A - B\|_F^2 \leq C\, E_{L,k}(A)?$$

The exponent and approximation constant must be independent of k, L, A. A deterministic algorithm is allowed. The output must retain rank at most k.

Reference

Noah Amsel, Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson, *Quasi-optimal Hierarchically Semi-separable Matrix Approximation*, SIAM Journal on Matrix Analysis and Applications **47** (2026), 586–621, §3.4; Theorems 6 and 12. That section expressly asks for a polynomial-time constant-factor approximation.

Status evidence

The known squared-error factor is $O(L)$. The lower bound near 2 concerns the analyzed greedy algorithm, not all algorithms. Searches for "HSS constant-factor approximation 2026" and "hierarchically semiseparable approximation constant 2026" found no resolution. The latest arXiv version is v2, September 6, 2025; the journal version appeared online May 8, 2026.

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Rechecked Amsel et al., §3.4 and Appendix B.1 and its 2026 journal record. Constant-factor polynomial-time HSS approximation remains unresolved. Constant-factor and later HSS searches found no solution; the known depth-dependent factor is weaker than this target.